

Projects Portfolio

Embedded Systems · Aerospace · Automotive · Cybersecurity · Medical

E2-CAD

EV Charging Station / V2G

2025 - 2026

Reverse engineering and analysis of embedded code on 3 boards (Microchip PIC, STM32). Test bench design and setup. Charging station debugging.

ISO 15118

OCPP

PLC

HomePlug

Linux

Python

Details

V2G (Vehicle-to-Grid) project. ISO 15118 and OCPP 1.6/2.0 protocols. PLC HomePlug Green PHY communication analysis between vehicle and charging station. Development of diagnostic tools in Python on Linux.

SAFRAN

SSPC A350 — Solid State Power Controller

2024 - 2025

Low-level driver development for the A350 program power controller. On-target validation testing.

C

Lauterbach TRACE32

DO-178

Details

DO-178C DAL-A context. C development on real-time microcontroller. Advanced debugging with Lauterbach TRACE32 probe. Unit and validation tests on target board.

THALES

Defense Radio LPM + N-MMR

2022 - 2023

Application layer development under embedded Linux for 2 military radio projects. HLT testing.

Linux

C/C++

Websockets

Bash

Details

Two defense radio projects: LPM (Military Programming Law) and N-MMR. C/C++ development on embedded Linux. Real-time Websockets communication. Bash scripts for HLT (High Level Tests) automation.

VALEO

Valeo Egypt Audit

2022

Audit of unit and integration testing processes on-site in Egypt. Improvement proposals.

NI Mastre

Vector CANoe

Details

Audit mission at the Valeo site in Cairo. Evaluation of test tools (NI Mastre, Vector CANoe). Quality process analysis and improvement proposals for validation teams.

ZODIAC / SAFRAN

GENOME / CORAC — SSPC A350

2017 - 2020

Low-level drivers, uAFDX to CAN gateway, diagnostics. Flight test campaign at UBC Vancouver, Canada.

Microchip dsPIC

CAN

RS485

DO-178

Details

CORAC/GENOME research program. Microchip dsPIC driver development. uAFDX to CAN bus communication gateway. RS485 protocol. Flight test campaign at the University of British Columbia (UBC), Vancouver.

ENEDIS

Linky Cybersecurity

2015 - 2016

Penetration testing and security audit of Linky data concentrators. Intrusion tests on embedded Linux systems.

Kali Linux

Nmap

Wireshark

Scapy

SDR

Python

Details

Penetration tests on Linky smart meter data concentrators. Network analysis with Nmap, Wireshark, Scapy. SDR radio. System security audit under Kali Linux. Vulnerability report.

SORIN

Platinum — BLE Pacemaker

2014 - 2015

Feasibility study for Bluetooth Smart integration in a cardiac implant. Smartphone protocol.

Cortex M0

BLE

Android

IEC 62304

Details

IEC 62304 context (medical software). Feasibility study for Bluetooth Low Energy integration in a pacemaker. Cortex M0 firmware development and Android communication application.

ASSOCIATION

Workshop-DIY

2020 - present

Founder. 18+ workshops for youth. STEM education. Arduino, ESP32, drone, and AI projects.

Arduino

ESP32

AI

3D Printing

Robotics

Details

Founder of the Workshop-DIY association. Organized over 18 educational workshops for young people (ages 8-18). Hands-on projects: Arduino, ESP32, 3D printing, drones, artificial intelligence, robotics.

Abdelhak Bourdim — Embedded Systems Engineer

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